



Universidade do Minho
Departamento de Informática

Exploração de dados com KNIME

LEI @ 2025/2026, 2º sem

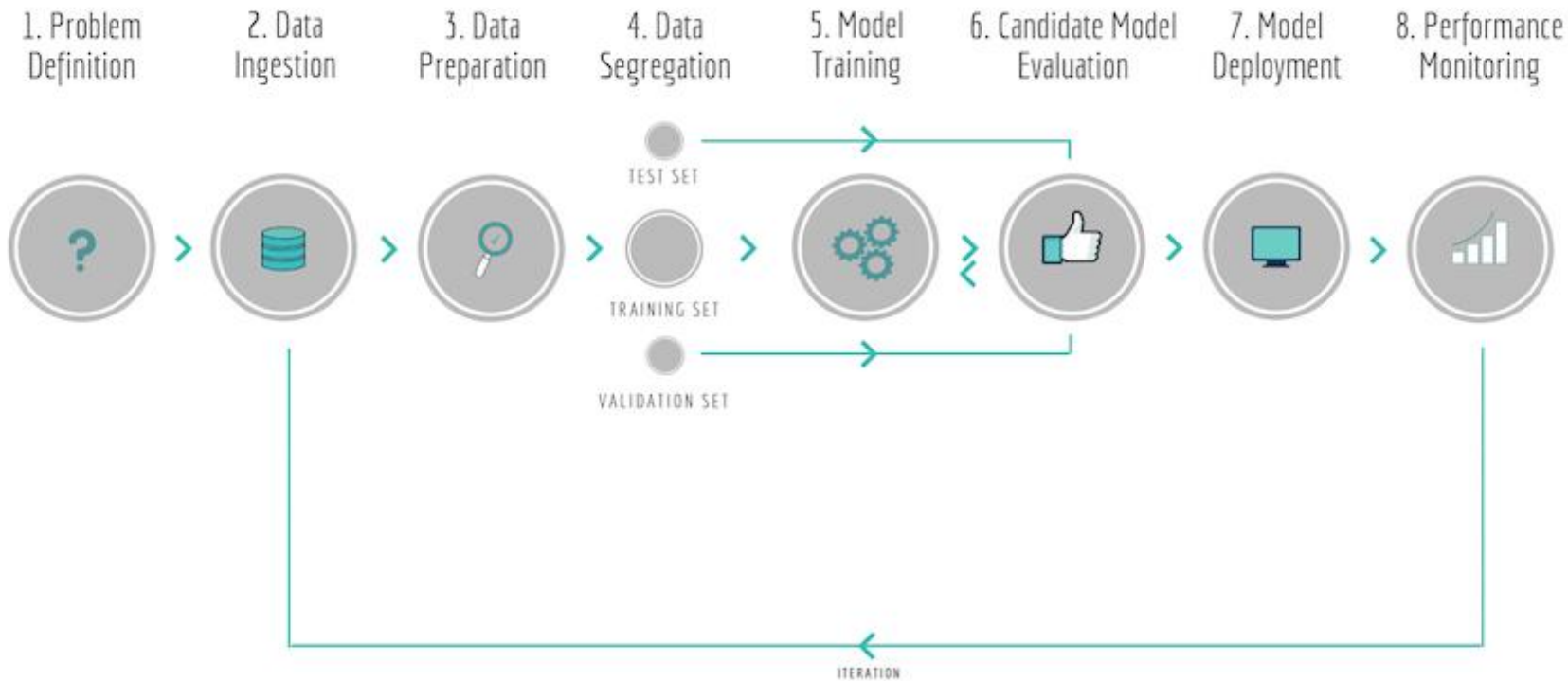


- KNIME
 - Bons hábitos
 - Metanodos
 - Ingestão de dados
 - Partição de dados
- Qualidade dos dados
- Exploração de dados
- Experimentação
(*hands on*)





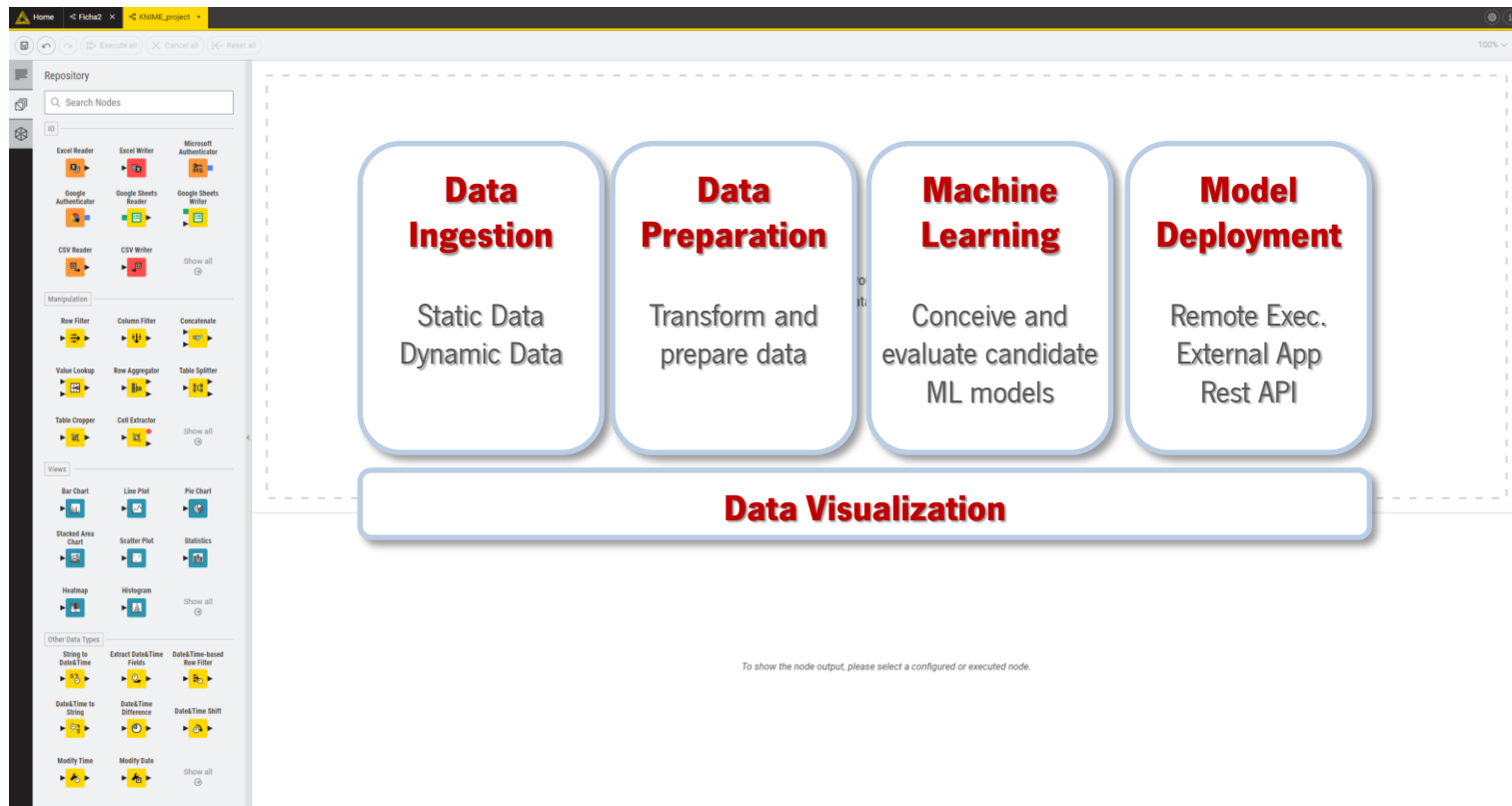
A Machine Learning Pipeline

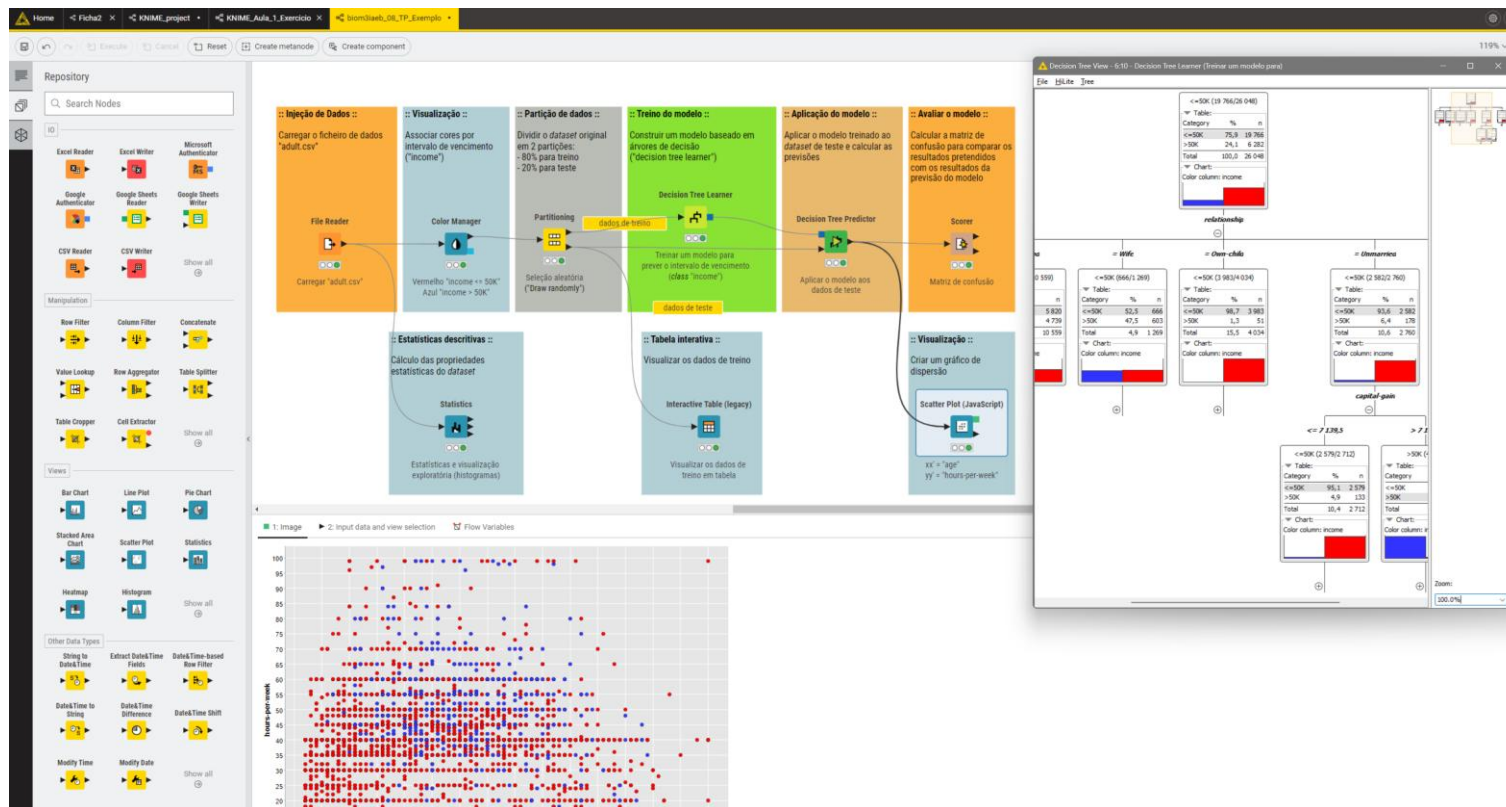


(<https://towardsdatascience.com/architecting-a-machine-learning-pipeline-a847f094d1c7>)



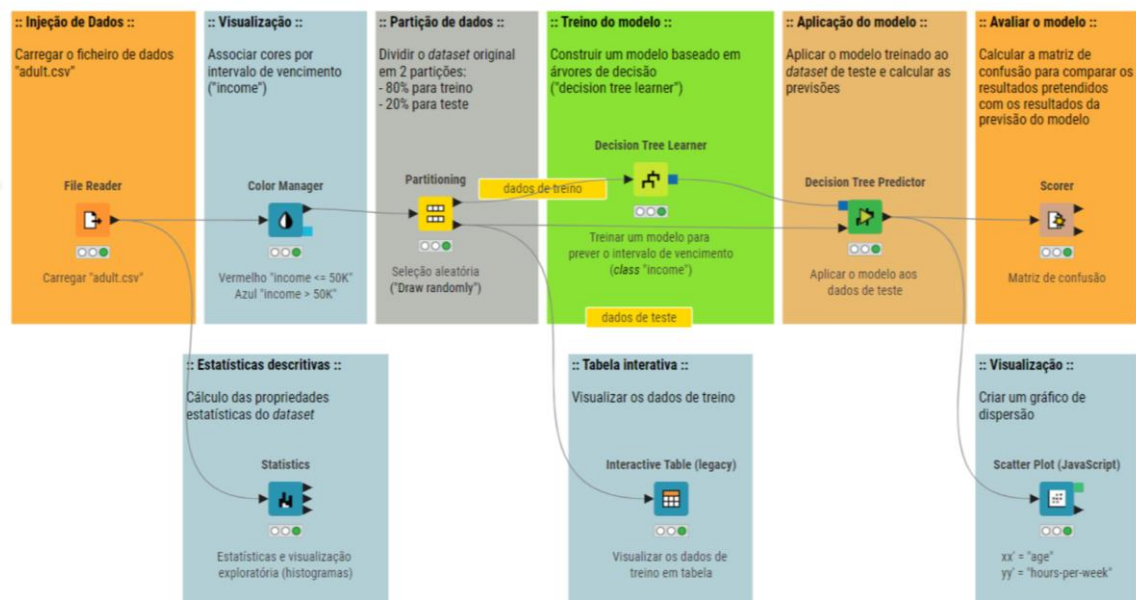
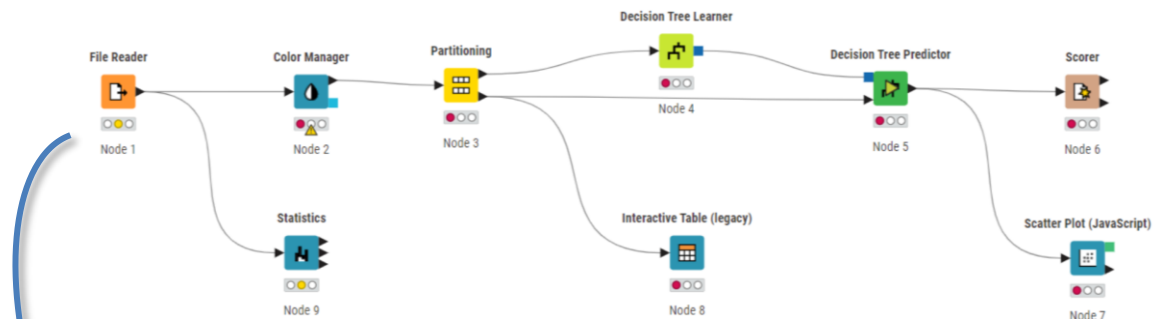
Fluxo de Trabalho Típico @ Knime

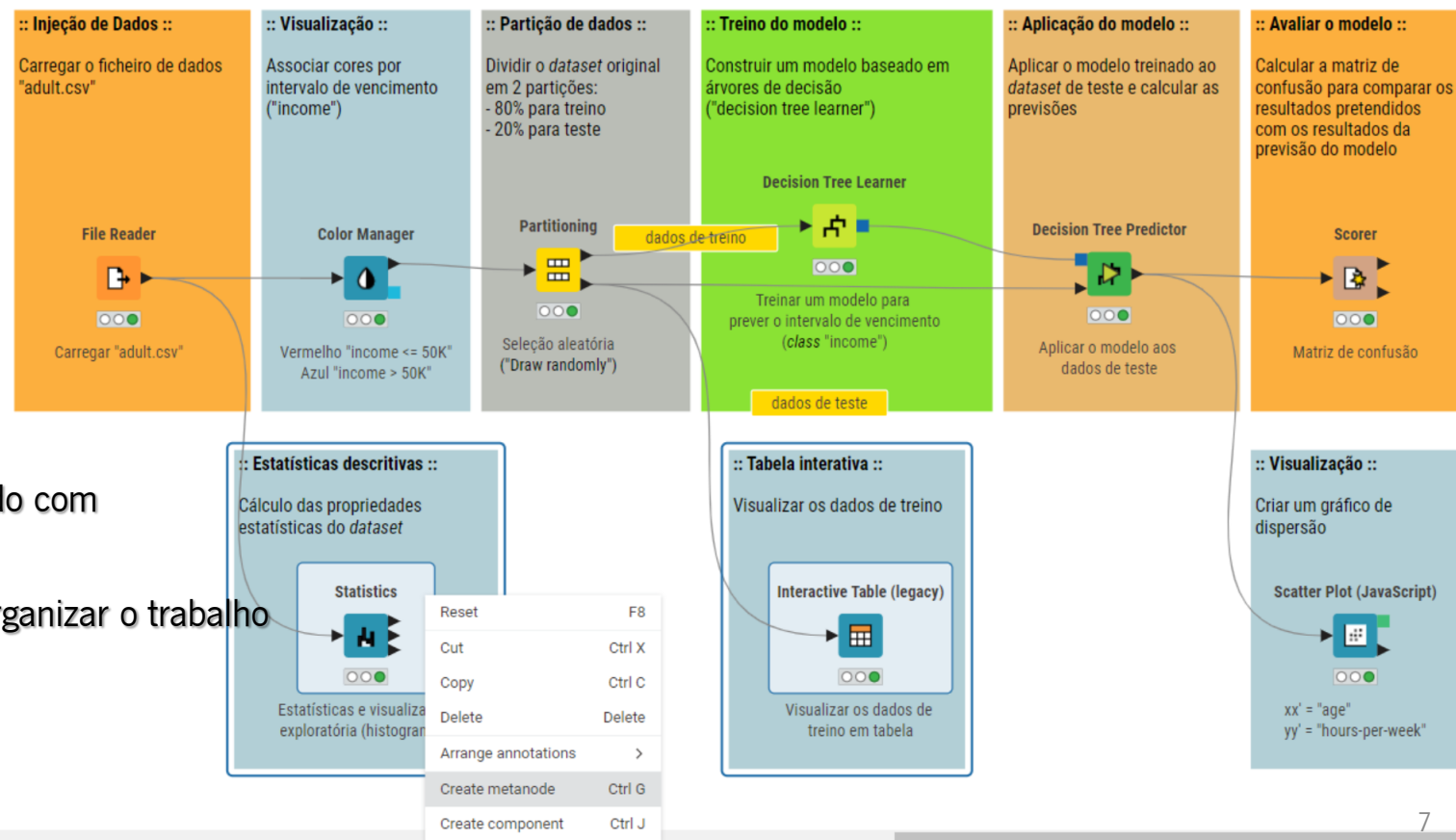




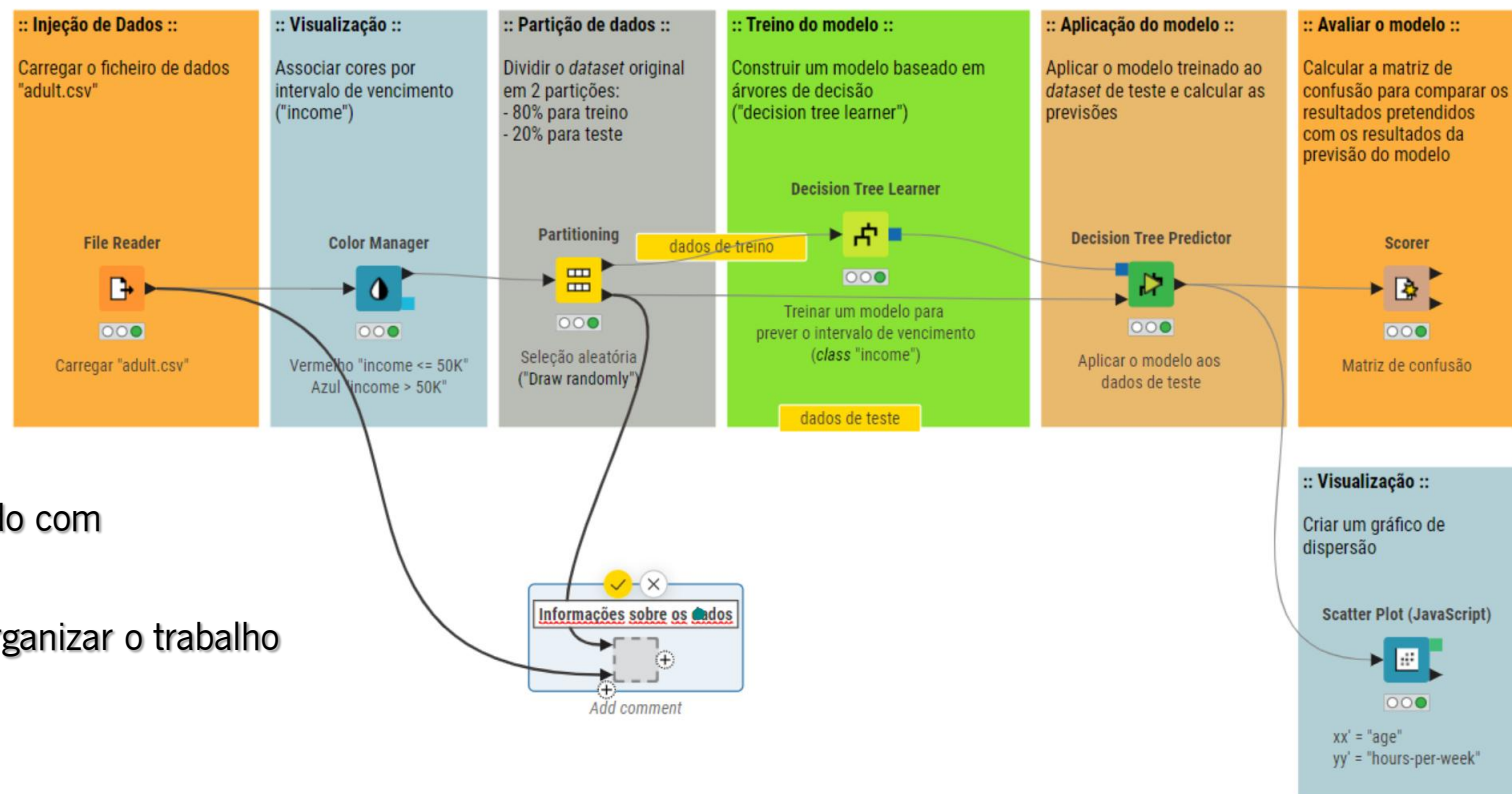


- Dar novos nomes aos nodos
- Adicionar anotações
- Utilizar metanodos

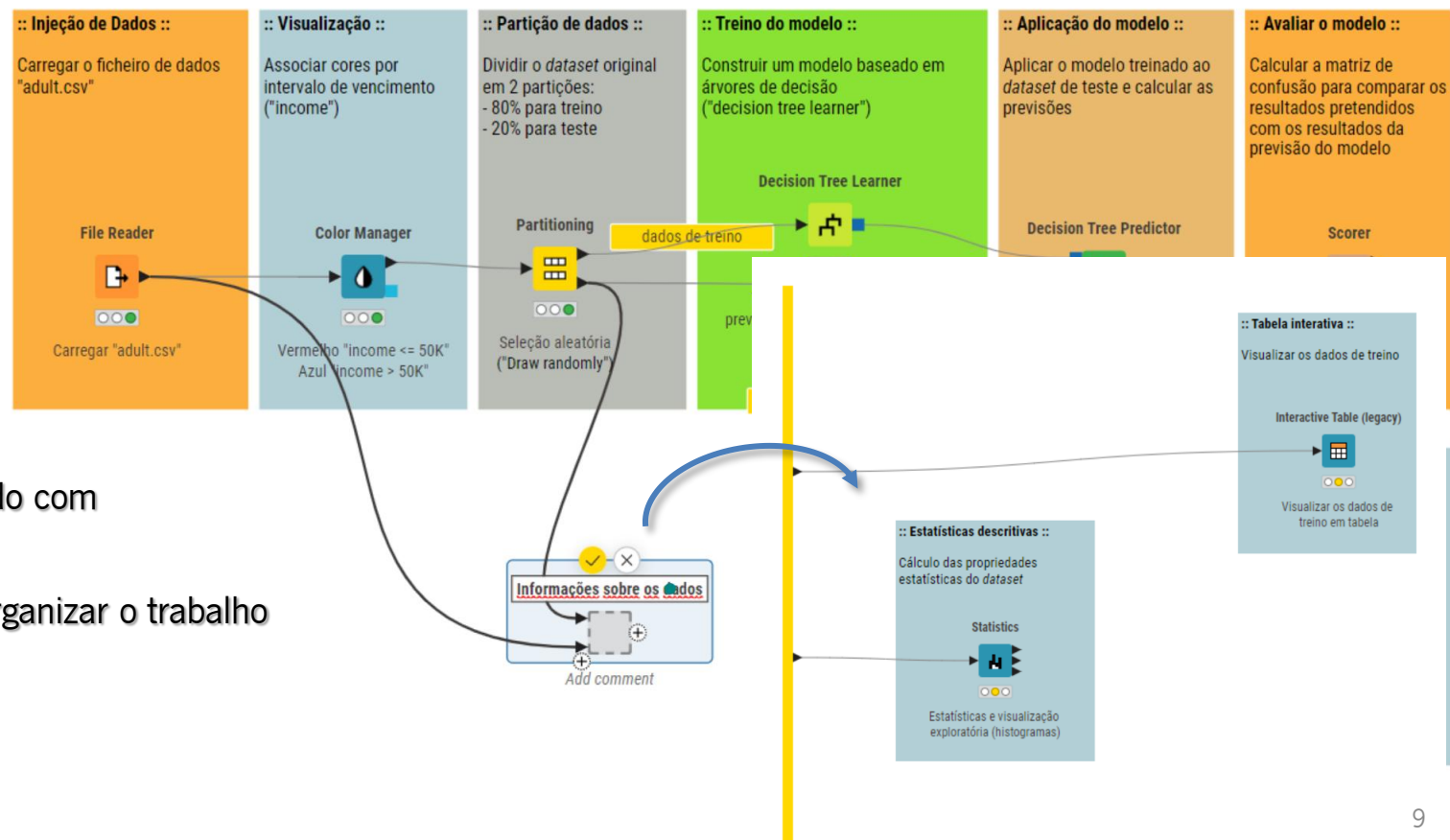




- Um metanodo é um nodo com outros nodos dentro
- Usar metanodos para organizar o trabalho



- Um metanodo é um nodo com outros nodos dentro
- Usar metanodos para organizar o trabalho

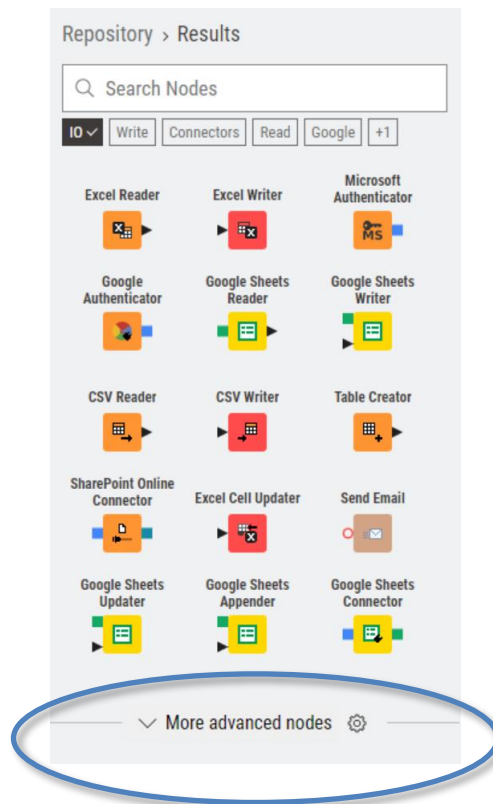


- Um metanodo é um nodo com outros nodos dentro
- Usar metanodos para organizar o trabalho



Ingestão de dados

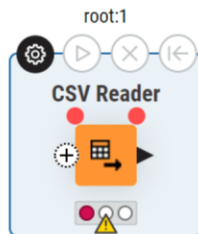
KNIME Data Readers





Ingestão de dados

KNIME Data Readers



Add comment

Dialog - 4:1 - CSV Reader

File

Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from: Local File System

Mode: ☒ File ☐ Files in folder

File: G:\O meu disco\2023 2024\15 lebiom3aeb2324\aulas\biom3aeb 08\ptanic_dataset.csv Browse...

Reader options

Format

Autodetect format

Column delimiter: , Row delimiter: ☒ Line break ☐ Custom \r\n

Quote char: " Quote escape char: \

Comment char: #

☒ Has column header ☐ Has row ID

☐ Support short data rows ☐ Prepend file index to row ID

Preview

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	T	Passen...	I	Survived	T	Pclass	S	Name	S	Sex	D	Age	I	SibSp	I	Parch	S	Ticket	D	Fare	S	Cabin	S
Row0	1	0	1	0	3			Braund, Mr. Owen Harris	male	22			1	0				A/5 21171	7.25	?		S	
Row1	2	1	1	1	1			Cummings, Mrs. John Br...	female	38			1	0				PC 17599	71.283	?		C	
Row2	3	1	1	1	3			Hekkinen, Miss. Laina	female	26			0	0				STON/O2. 3...	7.925	?		S	
Row3	4	1	1	1	1			Putrelle, Mrs. Jacques ...	female	35			1	0				113803	53.1	?		C123	S
Row4	5	0	3	0	3			Allen, Mr. William Henry	male	35			0	0				373450	8.05	?			S
Row5	6	0	3	0	3			Moran, Mr. James	male	?			0	0				330877	8.458	?			Q
Row6	7	0	1	0	1			McCarthy, Mr. Timothy J	male	54			0	0				17463	51.862	?			S
Row7	8	0	3	0	3			Palsson, Master. Gosta...	male	2			3	1				349909	21.075	?			S
Row8	9	1	3	1	3			Johnson, Mrs. Oscar ...	female	27			0	2				347742	11.133	?			S
Row9	10	1	2	1	2			Nasser, Mrs. Nicholas (...)	female	14			1	0				237736	30.071	?			C
Row10	11	1	3	1	3			Sandstrom, Miss. Marg...	female	4			1	1				PP 9549	16.7	?		IG6	S

► 1: File Table

Flow Variables



Ingestão de dados

KNIME Data Readers



Add comment

Dialog - 4:1 - Excel Reader

File and Sheet Data Area Advanced Transformation Flow Variables Job Manager Selection Memory Policy

Input Location

Read from Local File System

Mode ☒ File ☐ Files in folder

File G:\O meu disco\2023 2024\15 lebiom3aeb2324\aulas\biom3aeb 08\titanic_dataset.xlsx Browse...

Select Sheet

☒ First with data (titanic_dataset)

☐ By name titanic_dataset

☐ By position 0 (Position start with 0,)

Preview File Content

Preview with current settings

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	T	Passen...	I	Survived	T	Pclass	S	Name	S	Sex	D	Age	I	SibSp	I	Parch	S	Ticket	D	Fare	S	Cabin
Row0	1	0	3					Braund, Mr. Owen Harris	male	22	1	0		0			A/5 21171	7.25	?			
Row1	2	1	1					Cumings, Mrs. John Br...	female	38	1	0		0			PC 17599	71.283	?		C85	
Row2	3	1	3					Heikinen, Miss. Laina	female	26	0	0		0			STON/O2. 3...	7.925	?			
Row3	4	1	1					Futrelle, Mrs. Jacques ...	female	35	1	0		0			113803	53.1	?		C123	
Row4	5	0	3					Allen, Mr. William Henry	male	35	0	0		0			373450	8.05	?			
Row5	6	0	3					Moran, Mr. James	male	?	0	0		0			330877	8.458	?			
Row6	7	0	1					McCarthy, Mr. Timothy J	male	54	0	0		0			17463	51.862	?		E46	
Row7	8	0	3					Palsson, Master. Gosta...	male	2	3	1		1			349909	21.075	?			
Row8	9	1	3					Johnson, Mrs. Oscar ...	female	27	0	2		2			347742	11.133	?			
Row9	10	1	2					Nasser, Mrs. Nicholas (...)	female	14	1	0		0			237736	30.071	?			
Row10	11	1	3					Sandstrom, Miss. Marg...	female	4	1	1		1			PP 9549	16.7	?		G6	
Row11	12	1	1					Bonnell, Miss. Elizabeth	female	58	0	0		0			113783	26.55	?		C103	
Row12	13	0	3					Saunders, Mr. Willa...	male	20	0	0		0			A/5. 2151	8.05	?			
Row13	14	0	3					Andersson, Mr. Anders...	male	39	1	5		5			347082	31.275	?			
Row14	15	0	3					Vestrom, Miss. Hulda A...	female	14	0	0		0			350406	7.854	?			
Row15	16	1	2					Hewlett, Mrs. (Mary D ...)	female	55	0	0		0			248706	16	?			
Row16	17	0	3					Rice, Master. Eugene	male	2	4	1		1			382652	29.125	?			
Row17	18	1	2					Williams, Mr. Charles E...	male	?	?	0		0			244373	13	?			
Row18	19	0	3					Vander Planke, Mrs. Ju...	female	31	1	0		0			345763	18	?			
Row19	20	1	3					Masseimari, Mrs. Patma	female	?	0	0		0			2649	7.225	?			

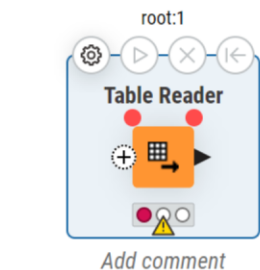
► 1: File Table

Flow Variables



Ingestão de dados

KNIME Data Readers



Dialog - 4:1 - Table Reader

Settings Transformation Advanced Settings Flow Variables Job Manager Selection Memory Policy

Input location

Read from: Local File System

Mode: ☒ File ☐ Files in folder

File: G:\D meu disco\2023 2024\15 lebiom3aeb2324\aulas\biom3aeb 09\Users_Sentiment.table

Row ID handling

☐ Use existing row ID ☐ Prepend table index to row ID

Preview

Row ID	I Custom...	I WebAc...	S Sentiment...	I Sentim...	S Marital...	S Gender	I Estim...	I Number...	I Age	I Target	I Availabl...	I Custr
Row0	11000	0	Slightly Negative	2	M	M	90000	0	44	1	1	1
Row1	11000	0	Slightly Negative	2	M	M	90000	0	44	1	1	1
Row2	11001	3	Slightly Positive	3	S	M	60000	1	45	1	0	1
Row3	11001	3	Slightly Positive	3	S	M	60000	1	45	1	0	1
Row4	11002	3	Slightly Positive	3	M	M	60000	1	45	1	1	1
Row5	11002	3	Slightly Positive	3	M	M	60000	1	45	1	1	1
Row6	11003	0	Very Negative	0	S	F	70000	1	42	1	0	1
Row7	11003	0	Very Negative	0	S	F	70000	1	42	1	0	1
Row8	11004	5	Very Positive	5	S	F	80000	4	42	1	1	1
Row9	11004	5	Very Positive	5	S	F	80000	4	42	1	1	1
Row10	11005	0	Very Negative	0	S	M	70000	1	45	1	1	1
Row11	11005	0	Very Negative	0	S	M	70000	1	45	1	1	1
Row12	11006	0	Very Negative	0	S	F	70000	1	44	1	1	1
Row13	11006	0	Very Negative	0	S	F	70000	1	44	1	1	1
Row14	11007	3	Slightly Positive	3	M	M	60000	2	46	1	1	1
Row15	11007	3	Slightly Positive	3	M	M	60000	2	46	1	1	1
Row16	11008	4	Positive	4	S	F	60000	3	46	1	1	1
Row17	11008	4	Positive	4	S	F	60000	3	46	1	1	1
Row18	11009	0	Very Negative	0	S	M	70000	1	46	1	0	1
Row19	11009	0	Very Negative	0	S	M	70000	1	46	1	0	1
Row20	11010	0	Very Negative	0	S	F	70000	1	46	1	0	1
Row21	11010	0	Very Negative	0	S	F	70000	1	46	1	0	1
Row22	11011	4	Positive	4	M	M	60000	4	46	1	1	1
Row23	11011	4	Positive	4	M	M	60000	4	46	1	1	1

► 1: Read table ☒ Flow Variables



Dados Carregados

► 1: File Table  Flow Variables

Rows: 891 | Columns: 12

Table Statistics

#	Row...	PassengerId Number (integer)	Survived Number (integer)	Pclass Number (integer)	Name String	Sex String	Age Number (double)	SibSp Number (integer)	Parch Number (integer)	Ticket String	Fare Number (double)	Cabin String	Embarked String
1	Row0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	🔴	S
2	Row1	2	1	1	Cumings, Mrs. John Brad...	female	38	1	0	PC 17599	71.283	C85	C
3	Row2	3	1	3	Heikinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	🔴	S
4	Row3	4	1	1	Futrelle, Mrs. Jacques He...	female	35	1	0	113803	53.1	C123	S
5	Row4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	🔴	S
6	Row5	6	0	3	Moran, Mr. James	male	🔴	0	0	330877	8.458	🔴	Q
7	Row6	7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.862	E46	S
8	Row7	8	0	3	Palsson, Master. Gosta L...	male	2	3	1	349909	21.075	🔴	S
9	Row8	9	1	3	Johnson, Mrs. Oscar W (...)	female	27	0	2	347742	11.133	🔴	S
10	Row9	10	1	2	Nasser, Mrs. Nicholas (A...	female	14	1	0	237736	30.071	🔴	C
11	Row10	11	1	3	Sandstrom, Miss. Margu...	female	4	1	1	PP 9549	16.7	G6	S
12	Row11	12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103	S
13	Row12	13	0	3	Saunderscock, Mr. William...	male	20	0	0	A/5. 2151	8.05	🔴	S
14	Row13	14	0	3	Andersson, Mr. Anders J...	male	39	1	5	347082	31.275	🔴	S
15	Row14	15	0	3	Vestrom, Miss. Hulda Am...	female	14	0	0	350406	7.854	🔴	S
16	Row15	16	1	2	Hewlett, Mrs. (Mary D Kin...	female	55	0	0	248706	16	🔴	S
17	Row16	17	0	3	Rice, Master. Eugene	male	2	4	1	382652	29.125	🔴	Q
18	Row17	18	1	2	Williams, Mr. Charles Eug...	male	🔴	0	0	244373	13	🔴	S
19	Row18	19	0	3	Vander Planke, Mrs. Juliu...	female	31	1	0	345763	18	🔴	S
20	Row19	20	1	3	Masselmani, Mrs. Fatima	female	🔴	0	0	2649	7.225	🔴	C
21	Row20	21	0	2	Fynney, Mr. Joseph J	male	35	0	0	239865	26	🔴	S
22	Row21	22	1	2	Beesley, Mr. Lawrence	male	34	0	0	248698	13	D56	S
23	Row22	23	1	3	McGowan, Miss. Anna "A..."	female	15	0	0	330923	8.029	🔴	Q
24	Row23	24	1	1	Sloper, Mr. William Thom...	male	28	0	0	113788	35.5	A6	S
25	Row24	25	0	3	Palsson, Miss. Torborg D...	female	8	3	1	349909	21.075	🔴	S
26	Row25	26	1	3	Asplund, Mrs. Carl Oscar ...	female	38	1	5	347077	31.387	🔴	S
27	Row26	27	0	3	Emir, Mr. Farred Chehab	male	🔴	0	0	2631	7.225	🔴	C
28	Row27	28	0	1	Fortune, Mr. Charles Alex...	male	19	3	2	19950	263	C23 C25 C27	S
29	Row28	29	1	3	O'Dwyer, Miss. Ellen "Nell..."	female	🔴	0	0	330959	7.879	🔴	Q
30	Row29	30	0	3	Todoroff, Mr. Lallo	male	🔴	0	0	349216	7.896	🔴	S
31	Row30	31	0	1	Uruchurtu. Don. Manuel E	male	40	0	0	PC 17601	27.721	🔴	C



Dados Carregados

Input features/Input vector

► 1: File Table

► Flow Variables

Rows: 891 | Columns: 12

Table Statistics

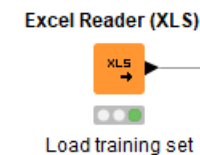
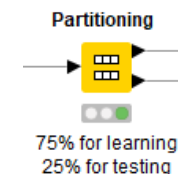
#	Row	PassengerId Number (integer)	Survived Number (integer)	Class Number (integer)	Name String	Sex String	Age Number (double)	SibSp Number (integer)	Parch Number (integer)	Ticket String	Fare Number (double)	Cabin String	Embarked String
1	Row0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	D	S
2	Row1	2	1	1	Cumings, Mrs. John Brad...	female	38	1	0	PC 17599	71.283	C85	C
3	Row2	3	1	3	Heikinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	D	S
4	Row3	4	1	1	Futrelle, Mrs. Jacques He...	female	35	1	0	113803	53.1	C123	S
5	Row4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	D	S
6	Row5	6	0	3	Moran, Mr. James	male	30	0	0	330877	8.458	D	Q
7	Row6	7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.862	E46	S
8	Row7	8	0	3	Palsson, Master. Gosta L...	male	2	3	1	349909	21.075	D	S
9	Row8	9	1	3	Johnson, Mrs. Oscar W (...)	female	27	0	2	347742	11.133	D	S
10	Row9	10	1	2	Nasser, Mrs. Nicholas (A...	female	14	1	0	237736	30.071	D	C
11	Row10	11	1	3	Sandstrom, Miss. Margu...	female	4	1	1	PP 9549	16.7	G6	S
12	Row11	12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103	S
13	Row12	13	0	3	Saunderscock, Mr. William...	male	20	0	0	A/5. 2151	8.05	D	S
14	Row13	14	0	3	Andersson, Mr. Anders J...	male	39	1	5	347082	31.275	D	S
15	Row14	15	0	3	Vestrom, Miss. Hulda Arn...	female	14	0	0	350406	7.854	D	S
16	Row15	16	1	2	Hewlett, Mrs. (Mary D Kin...	female	55	0	0	248706	16	D	S
17	Row16	17	0	3	Rice, Master. Eugene	male	2	4	1	382652	29.125	D	Q
18	Row17	18	1	2	Williams, Mr. Charles Eug...	male	30	0	0	244373	13	D	S
19	Row18	19	0	3	Vander Planke, Mrs. Juliu...	female	31	1	0	345763	18	D	S
20	Row19	20	1	3	Masselmani, Mrs. Fatima	female	26	0	0	2649	7.225	D	C
21	Row20	21	0	2	Fynney, Mr. Joseph J	male	35	0	0	239865	26	D	S
22	Row21	22	1	2	Beesley, Mr. Lawrence	male	34	0	0	248698	13	D56	S
23	Row22	23	1	3	McGowan, Miss. Anna "A..."	female	15	0	0	330923	8.029	D	Q
24	Row23	24	1	1	Sloper, Mr. William Thom...	male	28	0	0	113788	35.5	A6	S
25	Row24	25	0	3	Palsson, Miss. Torborg D...	female	8	3	1	349909	21.075	D	S
26	Row25	26	1	3	Asplund, Mrs. Carl Oscar ...	female	38	1	5	347077	31.387	D	S
27	Row26	27	0	3	Emir, Mr. Farred Chehab	male	26	0	0	2631	7.225	D	C
28	Row27	28	0	1	Fortune, Mr. Charles Alex...	male	19	3	2	19950	263	C23 C25 C27	S
29	Row28	29	1	3	O'Dwyer, Miss. Ellen "Nell..."	female	30	0	0	330959	7.879	D	Q
30	Row29	30	0	3	Todoroff, Mr. Lallo	male	26	0	0	349216	7.896	D	S
31	Row30	31	0	1	Uruchurtu, Don. Manuel E	male	40	0	0	PC 17601	27.721	D	C

Target/Class/Label



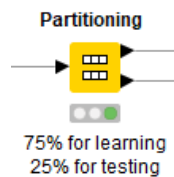
Partição de dados

- Construção de um modelo supervisionado de ML:
 - Usar um *dataset* de treino para treinar um modelo:
 - Aprendizagem
 - Usar um *dataset* de teste para testar o modelo:
 - Avaliar o modelo com dados “não vistos” durante o treino
 - Usar um *dataset* de validação (aconselhável)
 - Promove uma validação sem viés de um modelo sobre o *dataset* de treino enquanto se afinam os hiperparâmetros





Partição de dados



Dialog - 2:14 - Partitioning (75% for learning)

File

First partition | Flow Variables | Memory Policy

Choose size of first partition

☐ Absolute

☒ Relative[%]

☐ Take from top

☐ Linear sampling

☐ Draw randomly

☒ Stratified sampling

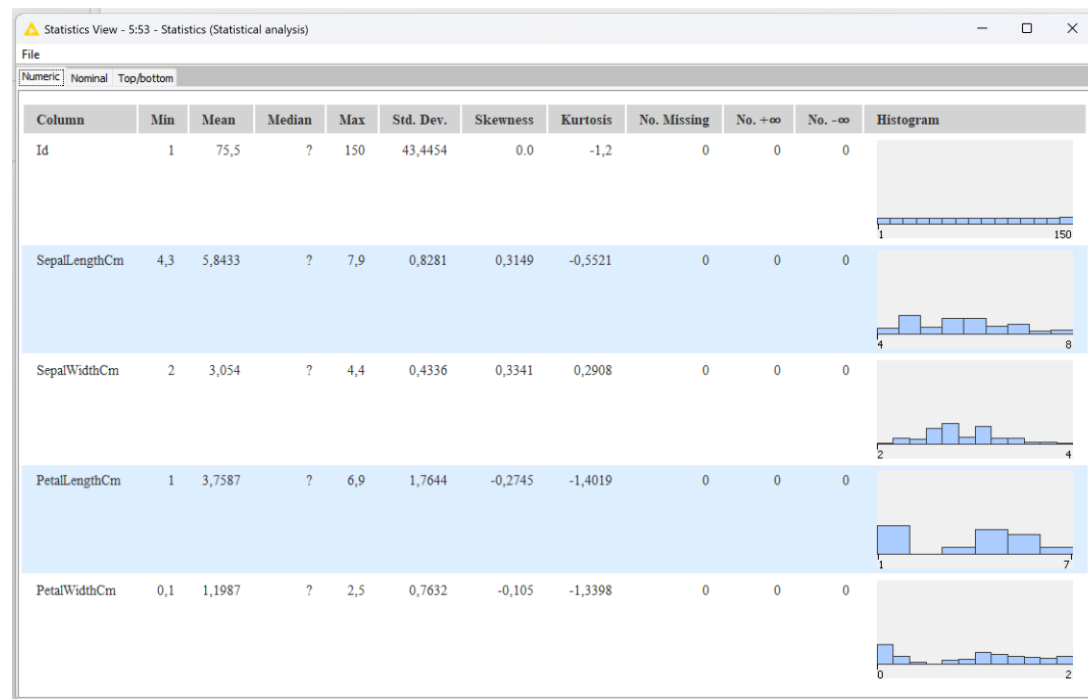
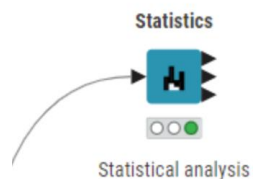
☒ Use random seed

OK Apply Cancel ?



Exploração dos dados

- Porquê?
 - Compreender os dados e as suas características;
 - Analisar a qualidade dos dados;
 - Perceber padrões e informações relevantes;

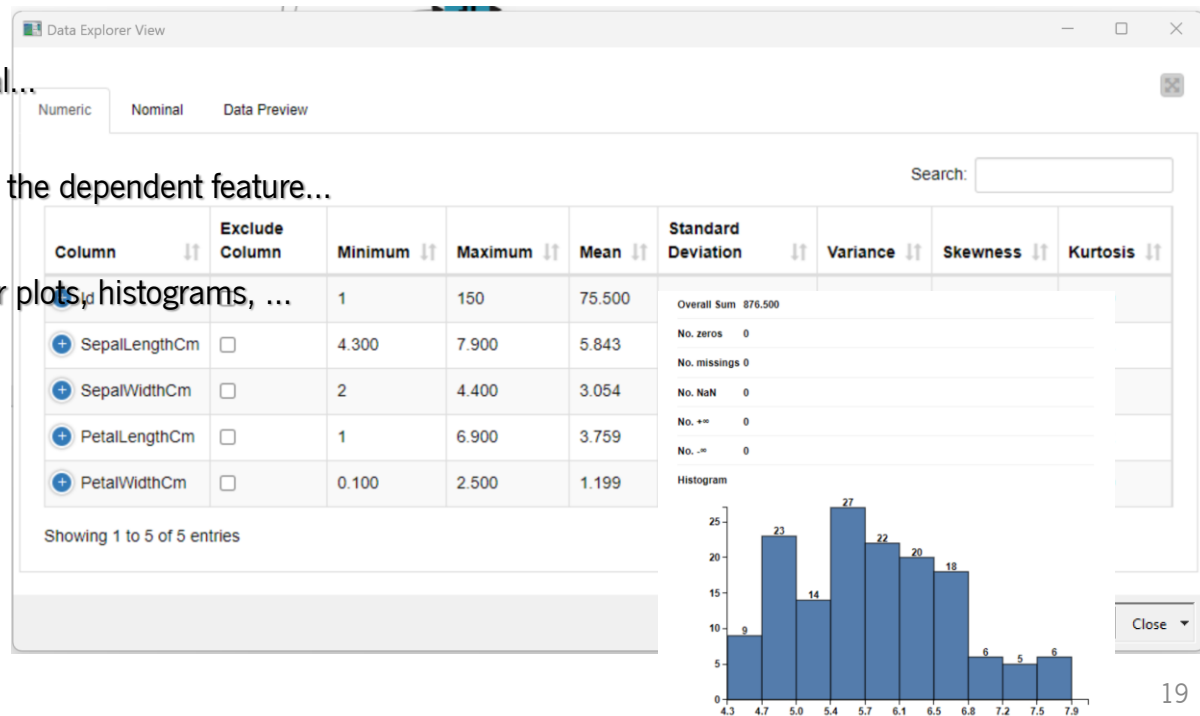




Exploração dos dados

■ Como?

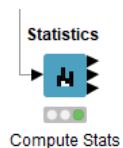
- Tendências centrais:
 - average, mode, median...
- Dispersão estatística
 - variance, standard deviation, interquartile range...
- Distribuição de probabilidades
 - Gaussian, Uniform, Exponential...
- Correlação/Dependência
 - between pairs of features, with the dependent feature...
- Visualização de dados
 - tables, charts, boxplots, scatter plots, histograms, ...





Exploração de dados

KNIME Statistics Node



Statistics Table - 0:726 - Statistics (Compute Stats)

File Hilite Navigation View

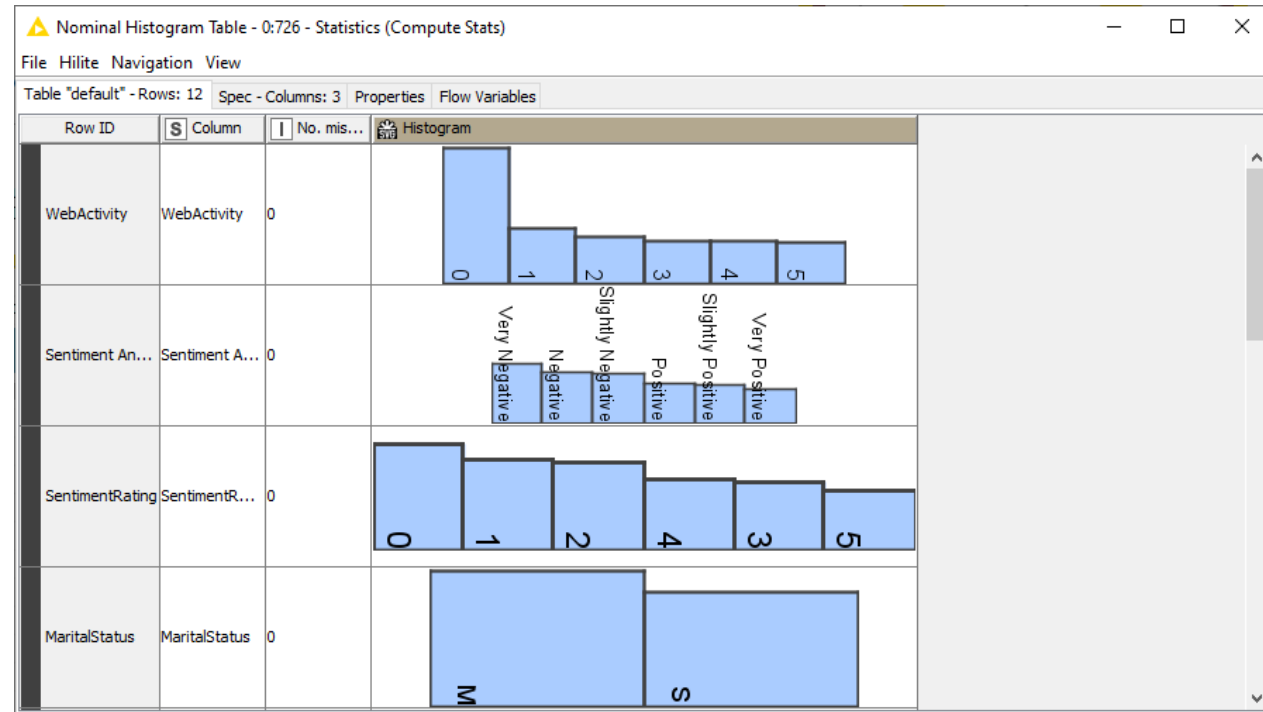
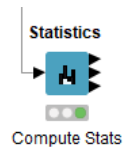
Table "default" - Rows: 11 Spec - Columns: 16 Properties Flow Variables

Row ID	S Column	D Min	D Max	D Mean	D Std. deviation	D Variance	D Skewness	D Kurtosis	D Overall sum	I No. missings	I No
EstimatedYea...	EstimatedYe...	10,000	170,000	57,718.072	32,091.91	1,029,890,7...	0.796	0.617	875,410,000	0	0
NumberOfCo...	NumberOfC...	0	4	1.465	1.145	1.311	0.43	-0.457	22,227	0	0
Age	Age	29	100	48.203	11.3	127.694	0.571	-0.182	731,101	0	0
Target	Target	0	1	0.487	0.5	0.25	0.053	-1.997	7,383	0	0
Available401K	Available401K	0	1	0.696	0.46	0.211	-0.854	-1.27	10,562	0	0



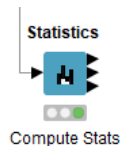
Exploração de dados

KNIME Statistics Node





Exploração de dados *KNIME Statistics Node*



Occurrences Table - 0:726 - Statistics (Compute Stats)

File Hilite Navigation View

Table "default" - Rows: 68 Spec - Columns: 36 Properties Flow Variables

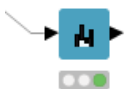
Row ID	I WebActivity	I Count (WebActivity)	D Relative Frequency (WebActivity)	S Sentiment Analysis	I Count (Sentiment Analysis)	D Relative Frequency (Sentiment Analysis)
Row0	0	9159	0.604	Very Negative	4173	0.275
Row1	1	1983	0.131	Negative	3122	0.206
Row2	2	1366	0.09	Slightly Negative	3023	0.199
Row3	3	963	0.063	Positive	1960	0.129
Row4	4	925	0.061	Slightly Positive	1690	0.111
Row5	5	771	0.051	Very Positive	1199	0.079
Row6	?	?	?	?	?	?
Row7	?	?	?	?	?	?
Row8	?	?	?	?	?	?
Row9	?	?	?	?	?	?
Row10	?	?	?	?	?	?
Row11	?	?	?	?	?	?
Row12	?	?	?	?	?	?
Row13	?	?	?	?	?	?
Row14	?	?	?	?	?	?
Row15	?	?	?	?	?	?
Row16	?	?	?	?	?	?
Row17	?	?	?	?	?	?
Row18	?	?	?	?	?	?
Row19	?	?	?	?	?	?
Row20	?	?	?	?	?	?
Row21	?	?	?	?	?	?



Exploração de dados

KNIME Data Explorer Node

Data Explorer



Data Explorer View

Numeric Nominal Data Preview

Search:

Column	Exclude Column	Minimum	Maximum	Mean	Standard Deviation	Variance	Skewness	Kurtosis	Overall Sum	No. zeros	No. missings
CustomerKey	☐	11000	27336	17559.847	5576.039	31092215.201	0.333	-1.566	266330201	0	0
WebActivity	☐	0	5	0.999	1.520	2.310	1.395	0.687	15159	9159	0
SentimentRating	☐	0	5	1.851	1.620	2.624	0.482	-0.958	28073	4173	0
EstimatedYearlyIncome	☐	10000	170000	57718.072	32091.910	1029890707.928	0.796	0.617	875410000	0	0
NumberOfContracts	☐	0	4	1.465	1.145	1.311	0.430	-0.457	22227	3711	0
Age	☐	29	100	48.203	11.300	127.694	0.571	-0.182	731101	0	0
Target	☐	0	1	0.487	0.500	0.250	0.053	-1.997	7383	7784	0
Available401K	☐	0	1	0.696	0.460	0.211	-0.854	-1.270	10562	4605	0
CustomerValueSegment	☐	1	3	2.097	0.689	0.475	-0.129	-0.898	31809	0	0
ChurnScore	☐	0	1	0.269	0.332	0.110	1.254	0.296	4078.300	5299	0
CallActivity	☐	1	5	3.237	1.262	1.594	-0.302	-0.915	49094	0	0

Showing 1 to 11 of 11 entries

Reset Apply Close

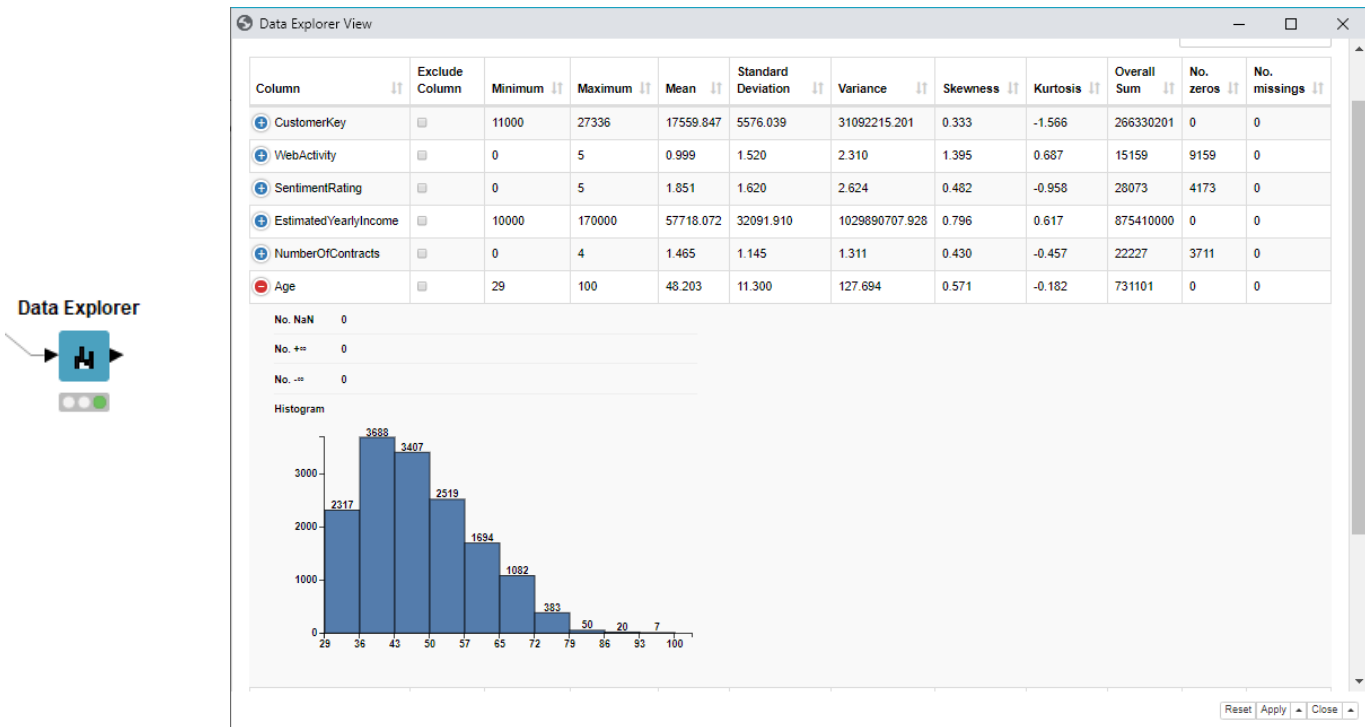


Install **KNIME JavaScript Views (Labs)** extension



Exploração de dados

KNIME Data Explorer Node



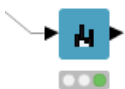
Install **KNIME JavaScript Views (Labs)** extension



Exploração de dados

KNIME Data Explorer Node

Data Explorer



Data Explorer View

Numeric Nominal Data Preview

Search:

Column	Exclude Column	No. missings	Unique values	All nominal values	Frequency Bar Chart
Sentiment Analysis	☐	0	6	Very Negative, Negative, Slightly Negative, Positive, Slightly Positive, Very Positive	
MaritalStatus	☐	0	2	M, S	
Gender	☐	0	2	M, F	
Products	☐	0	5	private investment, p+b investment, fund manager+, gold investment, co investment	

Showing 1 to 4 of 4 entries

Reset Apply Close

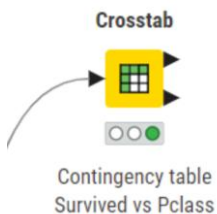


Install **KNIME JavaScript Views (Labs)** extension



Exploração de dados

KNIME Contingency Tables



Cross Tabulation of Survived by Pclass

Percent Row Percent	1	2	3	Total
0	8,26%	11,0361%	42,4605%	61,7566%
1	13,3751%	17,8704%	68,7546%	
	16,8302%	9,6128%	11,8004%	38,2434%
	44,0082%	25,1359%	30,8559%	
Total	25,0902%	20,6489%	54,2608%	100%

- ☐ Frequency
- ☐ Expected
- ☐ Deviation
- ☒ Percent
- ☒ Row Percent
- ☐ Column Percent
- ☐ Cell Chi-Square

Max rows:

10

Max columns:

10

Exploração de dados KNIME Co.

Statistics for Table of Survived by Pclass

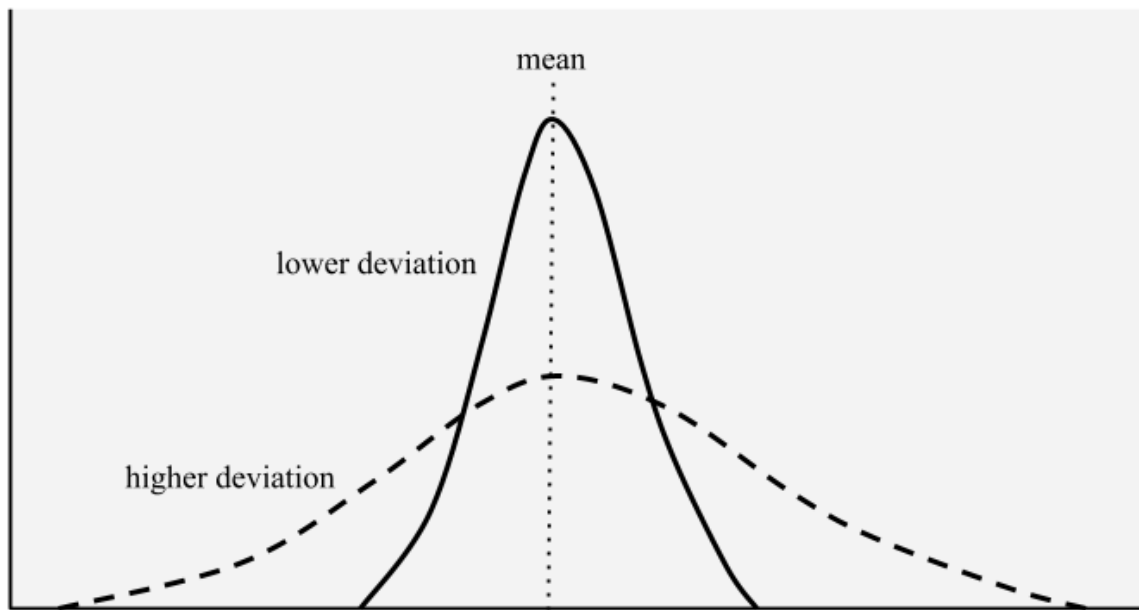
Statistic	DF	Value	Prob
Chi-Square	2	62 344,7717	0.0

Total sample size: 397386.0



Variância

- A variância mede a dispersão dos dados em relação à sua média;



Variance

Population

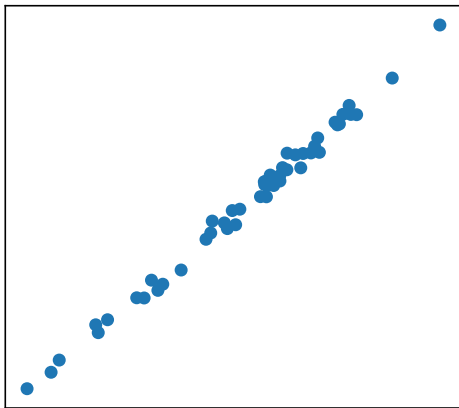
$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Sample

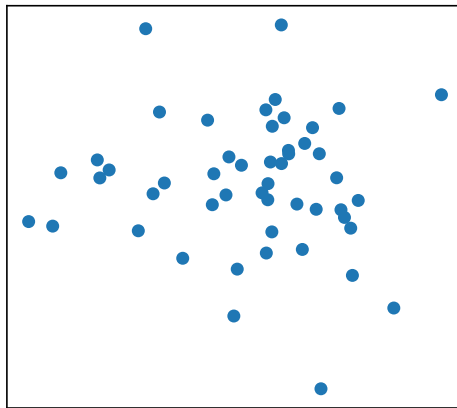
$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$



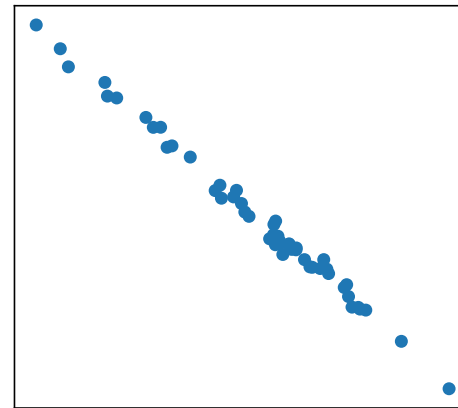
- A covariância mede quanto duas variáveis (atributos, colunas, *features*) dependem uma da outra;



alta covariância



covariância (quase) nula



alta covariância negativa



- A covariância mede quanto duas variáveis (atributos, colunas, *features*) dependem uma da outra;
- Calcular a covariância:
 - Pense nos dados das duas variáveis como vetores de alta dimensão;
 - Converta-os em vetores de variâncias da média;
 - Calcule o produto escalar (cosseno do ângulo entre eles) dos dois vetores;
 - Divida pelo tamanho da população;

For Population

$$\text{Cov}(x,y) = \frac{\sum (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$\text{Cov}(x,y) = \frac{\sum (x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$



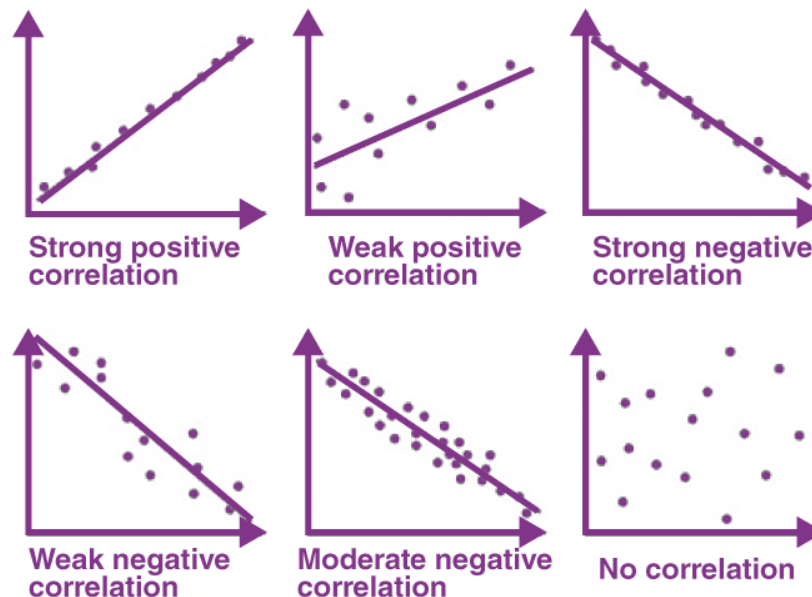
Covariância & Correlação

- Interpretar a **covariância** é difícil:
 - Baixa covariância (próxima de 0) significa que não há muita correlação entre as duas variáveis;
 - Alta covariância (muito superior a 0, ou negativa para relações inversas) significa que existe uma correlação entre as variáveis;
- Interpretar a **correlação** é mais fácil:
 - Valor de normalização da covariância dividido pelos desvios padrão de ambas as variáveis:
 - Correlação = -1: correlação inversa perfeita
 - Correlação = 0: sem correlação
 - Correlação = 1: correlação perfeita
- A **covariância** indica como duas variáveis mudam de valor, quando comparadas conjuntamente;
- Dependente da unidade de medida das variáveis;
- A **correlação** indica a proximidade do comportamento entre duas variáveis;
- Não depende da unidade de medida;

Tanto a **covariância** como a **correlação** medem o relacionamento e a dependência entre duas variáveis;

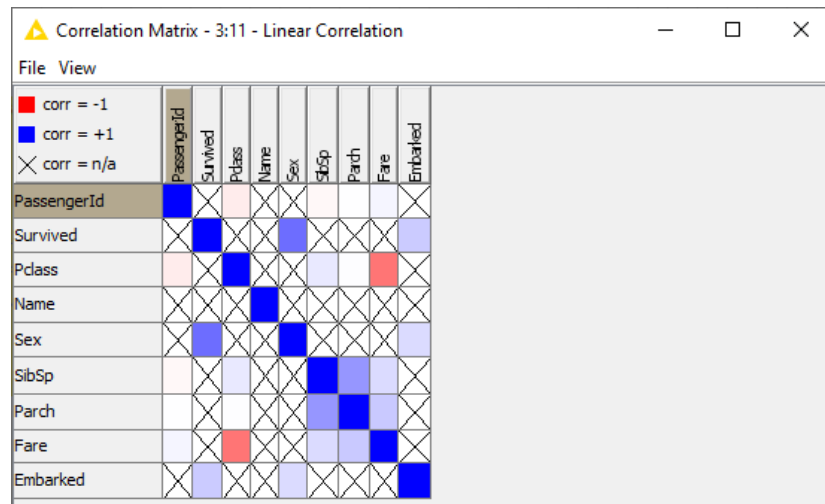
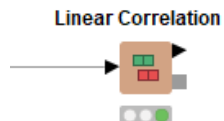


- Mas... a correlação não implica causalidade!!!
 - Apenas uma experiência controlada e aleatória pode fornecer informações sobre a causalidade;
 - Usar a correlação para decidir quais as experiências a realizar.





Matriz de Correlação



Correlation measure - 3:11 - Linear Correlation

File Hilite Navigation View

Table "Correlation values" - Rows: 9 Spec - Columns: 9 Properties Flow Variables

Row ID	D Passen...	D Survived	D Pclass	D Name	D Sex	D SibSp	D Parch	D Fare	D Embarked
PassengerId	1.0	?	-0.074684...	?	?	-0.02572993...	0.0026940469...	0.04019030952...	?
Survived	?	1.0	?	?	0.5737...	?	?	?	0.20559893...
Pclass	-0.0746846...	?	1.0	?	?	0.084898459...	0.0060928325...	-0.5393077410...	?
Name	?	?	?	1.0	?	?	?	?	?
Sex	?	0.57374697...	?	?	1.0	?	?	?	0.14153725...
SibSp	-0.0257299...	?	0.0848984...	?	?	1.0	0.4102760703...	0.14270976638...	?
Parch	0.00269404...	?	0.0060928...	?	?	0.410276070...	1.0	0.21318809003...	?
Fare	0.04019030...	?	-0.539307...	?	?	0.142709766...	0.2131880900...	1.0	?
Embarked	?	0.20559893...	?	?	0.1415...	?	?	?	1.0

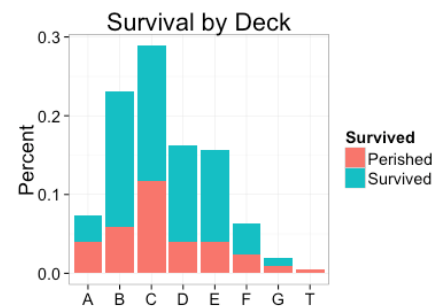
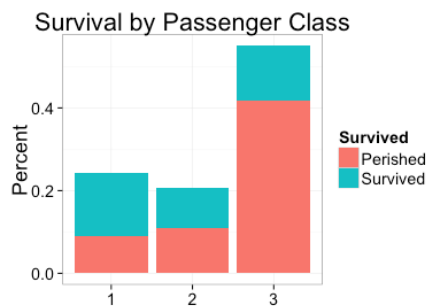
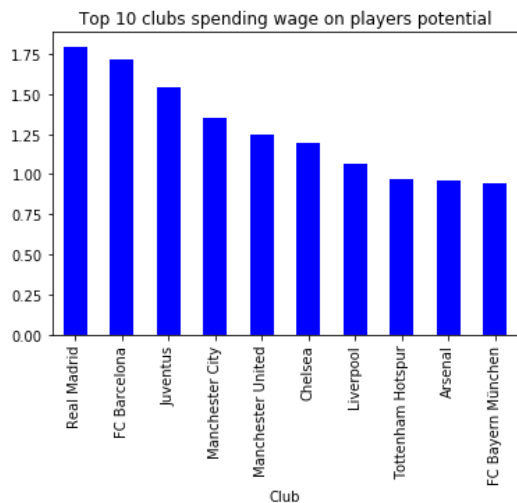
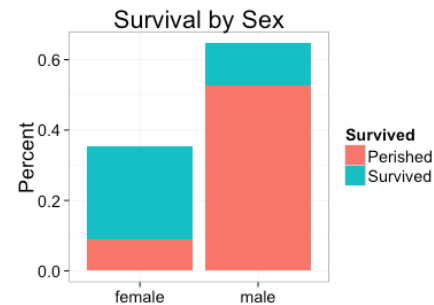
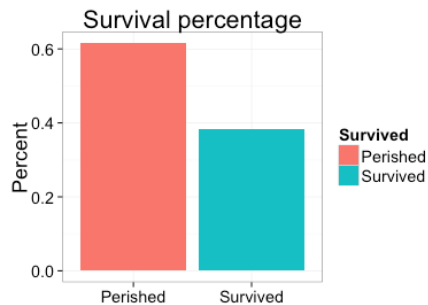
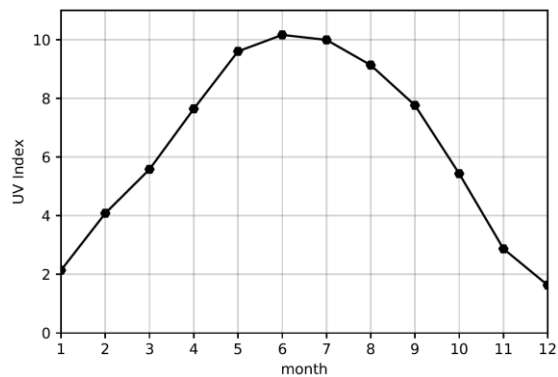


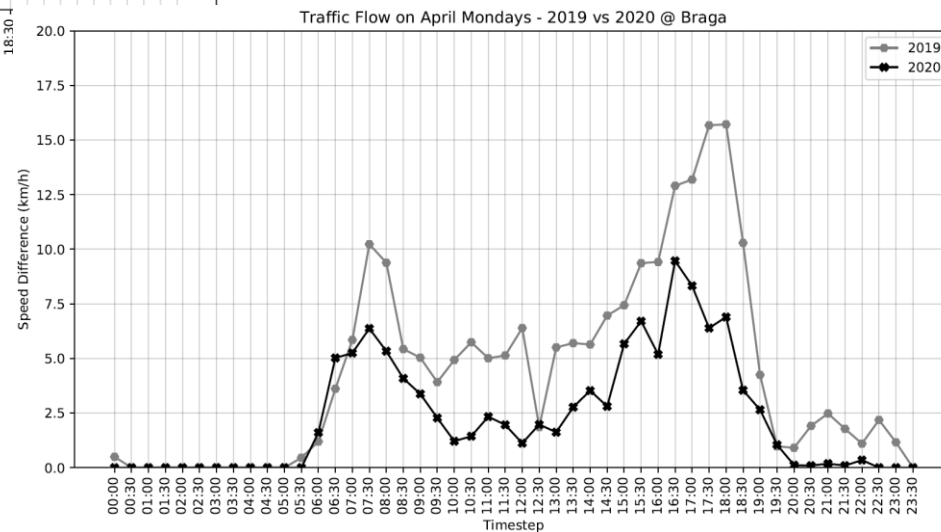
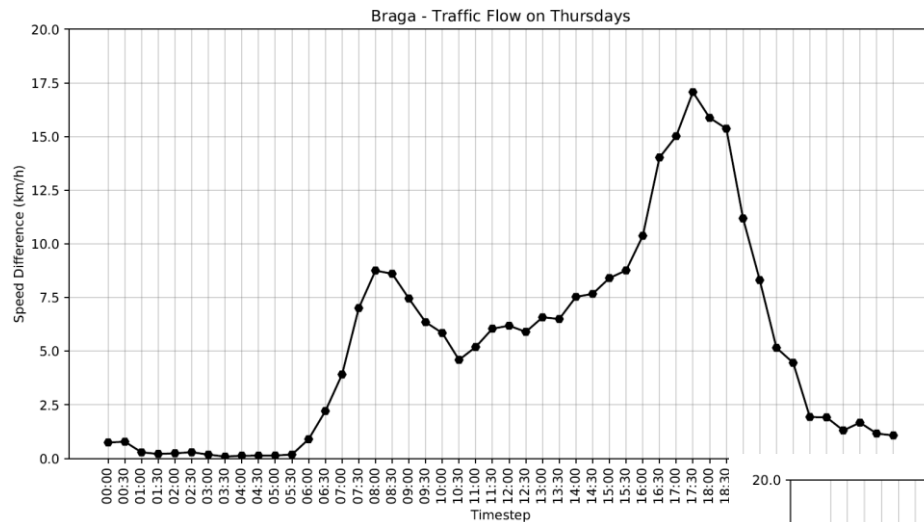
Matriz de Correlação

- Queremos manter atributos (*features*) altamente correlacionados?
- Ambos positiva e negativamente correlacionados?
- E quanto à correlação entre os atributos dependentes e independentes?
- ...



Visualização de Dados







Home

Building a Simple Classifier ×

Help

Preferences

Menu

Save

Undo

Redo

Execute all

Cancel all

Reset all

93%

Nodes

Search all nodes

IO

Excel Reader

Excel Writer

Microsoft Authenticator

CSV Reader

CSV Writer

Table Creator

SharePoint Online Connector

File Reader

Show all

Manipulation

Row Filter

Column Filter

Concatenate

Value Lookup

Row Aggregator

Table Splitter

String Cleaner

Table Cropper

Show all

Views

QUICK HANDS ON

To show the node output, please select a configured or executed node.



Universidade do Minho
Departamento de Informática

Exploração de dados com KNIME

LEI @ 2025/2026, 2º sem